

2026-02-28 - TRAFFIC ANALYSIS EXERCISE ANSWERS

Link to the exercise:

- <https://www.malware-traffic-analysis.net/2026/02/28/index.html>

Links to some tutorials I've written that should help with this exercise:

- <https://www.malware-traffic-analysis.net/tutorials.html>

ENVIRONMENT:

- LAN segment range: 10.2.28[.]0/24 (10.2.28[.]0 through 10.2.28[.]255)
- Domain: easyas123[.]tech
- AD environment name: EASYAS123
- Domain Controller: 10.2.28[.]2 – EASYAS123-DC
- LAN segment gateway: 10.2.28[.]1
- LAN segment broadcast address: 10.2.28[.]255

BACKGROUND:

As dynamic go-getter at a Security Operations Center (SOC), you check the Security Information and Event Management (SIEM) system and find several signature hits for NetSupport Manager RAT from 45.131.214[.]85 over TCP port 443. The activity started on 2026-02-28 at 19:55 UTC..

Using this information, you quickly retrieve a packet capture (pcap) of the traffic from the internal IP address that triggered these alerts. It's all on you now! You're expected to write up an incident report, so someone can track down the infected computer and put a stop to this nonsense!

Armed with pcap, you intend to find that infected host.

TASK:

For this exercise, answer the following questions for your incident report:

- What is the IP address of the infected Windows client?
- What is the MAC address of the infected Windows client?
- What is the host name of the infected Windows client?
- What is the user account name from the infected Windows client?
- What is the full name of the user from the user account?

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ANSWERS:

Victim Details:

- IP address: 10.1.28[.]88
- Host name: DESKTOP-TEYQ2NR
- MAC address: 00:19:d1:b2:4d:ad
- Windows user account name: brolf
- Full name of the user: Becka Rolf

NetSupport Manager RAT traffic on:

- 45.131.214[.]85:443

2026-02-28-traffic-analysis-exercise.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

(http.request or tls.handshake.type eq 1) and !ssdp and ip.addr eq 45.131.214.85 basic basic+ basic+dns

Time	Src	port	Dst	port	Host	Info
2026-02-28 19:55:51	10.2.28.88	51912	45.131.214.85	443	45.131.214.85	POST http://45.131.214.85/f
2026-02-28 19:55:51	10.2.28.88	51912	45.131.214.85	443	45.131.214.85	POST http://45.131.214.85/f
2026-02-28 19:55:52	10.2.28.88	51912	45.131.214.85	443	45.131.214.85	POST http://45.131.214.85/f
2026-02-28 19:55:52	10.2.28.88	51912	45.131.214.85	443	45.131.214.85	POST http://45.131.214.85/f
2026-02-28 19:56:52	10.2.28.88	51912	45.131.214.85	443	45.131.214.85	POST http://45.131.214.85/f
2026-02-28 19:57:52	10.2.28.88	51912	45.131.214.85	443	45.131.214.85	POST http://45.131.214.85/f
2026-02-28 19:58:52	10.2.28.88	51912	45.131.214.85	443	45.131.214.85	POST http://45.131.214.85/f
2026-02-28 19:59:52	10.2.28.88	51912	45.131.214.85	443	45.131.214.85	POST http://45.131.214.85/f
2026-02-28 20:00:52	10.2.28.88	51912	45.131.214.85	443	45.131.214.85	POST http://45.131.214.85/f
2026-02-28 20:01:52	10.2.28.88	51912	45.131.214.85	443	45.131.214.85	POST http://45.131.214.85/f
2026-02-28 20:02:52	10.2.28.88	51912	45.131.214.85	443	45.131.214.85	POST http://45.131.214.85/f
2026-02-28 20:03:53	10.2.28.88	51912	45.131.214.85	443	45.131.214.85	POST http://45.131.214.85/f
2026-02-28 20:04:53	10.2.28.88	51912	45.131.214.85	443	45.131.214.85	POST http://45.131.214.85/f
2026-02-28 20:05:53	10.2.28.88	51912	45.131.214.85	443	45.131.214.85	POST http://45.131.214.85/f
2026-02-28 20:06:53	10.2.28.88	51912	45.131.214.85	443	45.131.214.85	POST http://45.131.214.85/f
2026-02-28 20:07:53	10.2.28.88	51912	45.131.214.85	443	45.131.214.85	POST http://45.131.214.85/f
2026-02-28 20:08:53	10.2.28.88	51912	45.131.214.85	443	45.131.214.85	POST http://45.131.214.85/f
2026-02-28 20:09:54	10.2.28.88	51912	45.131.214.85	443	45.131.214.85	POST http://45.131.214.85/f
2026-02-28 20:10:54	10.2.28.88	51912	45.131.214.85	443	45.131.214.85	POST http://45.131.214.85/f

Frame 2638: 272 bytes on wire (2176 bits), 272 bytes captured (2176 bits)

Ethernet II, Src: IntelCor_b2:4d:ad (00:19:d1:b2:4d:ad), Dst: Cisco_41:ae:25 (00:05:00:41:ae:25)

Internet Protocol Version 4, Src: 10.2.28.88, Dst: 45.131.214.85

Transmission Control Protocol, Src Port: 51912, Dst Port: 443, Seq: 1, Ack: 1, Len: 218

Hypertext Transfer Protocol

HTML Form URL Encoded: application/x-www-form-urlencoded

2026-02-28-traffic-analysis-exercise.pcap Packets: 15512 · Displayed: 264 (1.7%) Profile: Default

Shown above: Filtering in Wireshark to show the traffic on 45.131.214[.]85.

HINTS:

As always, since this pcap was retrieved from the infected Windows host, we can find that by pivoting on which internal IP address was communicating with 145.131.214[.]85, which is 10.1.21[.]88. Which you can easily see in the above image as the Source IP address in the Src column.

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Then we can filter on `nbns` in Wireshark to find the host name and view the frame details to find the MAC address.

2026-02-28-traffic-analysis-exercise.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

nbns

Time	Src	port	Dst	port	Info
2026-02-28 19:55:11	10.2.28.88	137	10.2.28.255	137	Registration NB DESKTOP-TEYQ2NR<20>
2026-02-28 19:55:11	10.2.28.88	137	10.2.28.255	137	Registration NB DESKTOP-TEYQ2NR<00>
2026-02-28 19:55:11	10.2.28.88	137	10.2.28.255	137	Registration NB EASYAS123<00>
2026-02-28 19:55:12	10.2.28.88	137	10.2.28.255	137	Registration NB EASYAS123<00>
2026-02-28 19:55:12	10.2.28.88	137	10.2.28.255	137	Registration NB DESKTOP-TEYQ2NR<00>
2026-02-28 19:55:12	10.2.28.88	137	10.2.28.255	137	Registration NB DESKTOP-TEYQ2NR<20>
2026-02-28 19:55:12	10.2.28.88	137	10.2.28.255	137	Registration NB DESKTOP-TEYQ2NR<20>
2026-02-28 19:55:12	10.2.28.88	137	10.2.28.255	137	Registration NB DESKTOP-TEYQ2NR<00>
2026-02-28 19:55:12	10.2.28.88	137	10.2.28.255	137	Registration NB EASYAS123<00>
2026-02-28 19:55:13	10.2.28.88	137	10.2.28.255	137	Registration NB EASYAS123<00>
2026-02-28 19:55:13	10.2.28.88	137	10.2.28.255	137	Registration NB DESKTOP-TEYQ2NR<00>
2026-02-28 19:55:13	10.2.28.88	137	10.2.28.255	137	Registration NB DESKTOP-TEYQ2NR<20>
2026-02-28 19:55:14	10.2.28.88	137	10.2.28.255	137	Registration NB EASYAS123<1e>
2026-02-28 19:55:15	10.2.28.88	137	10.2.28.255	137	Registration NB EASYAS123<1e>

Frame 135: 110 bytes on wire (880 bits), 110 bytes captured (880 bits)
Ethernet II, Src: IntelCor_b2:4d:ad (00:19:d1:b2:4d:ad), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
Internet Protocol Version 4, Src: 10.2.28.88, Dst: 10.2.28.255
User Datagram Protocol, Src Port: 137, Dst Port: 137
NetBIOS Name Service

Shown above: Filtering in Wireshark on `nbns` to find the host name.

To find the Windows user account name, filter on `Kerberos.CNameString` and select any of the frames from the results. Then we can drill down into the frame details to find the `CNameString` value, which is `bro1f`.

kerberos.CNameString

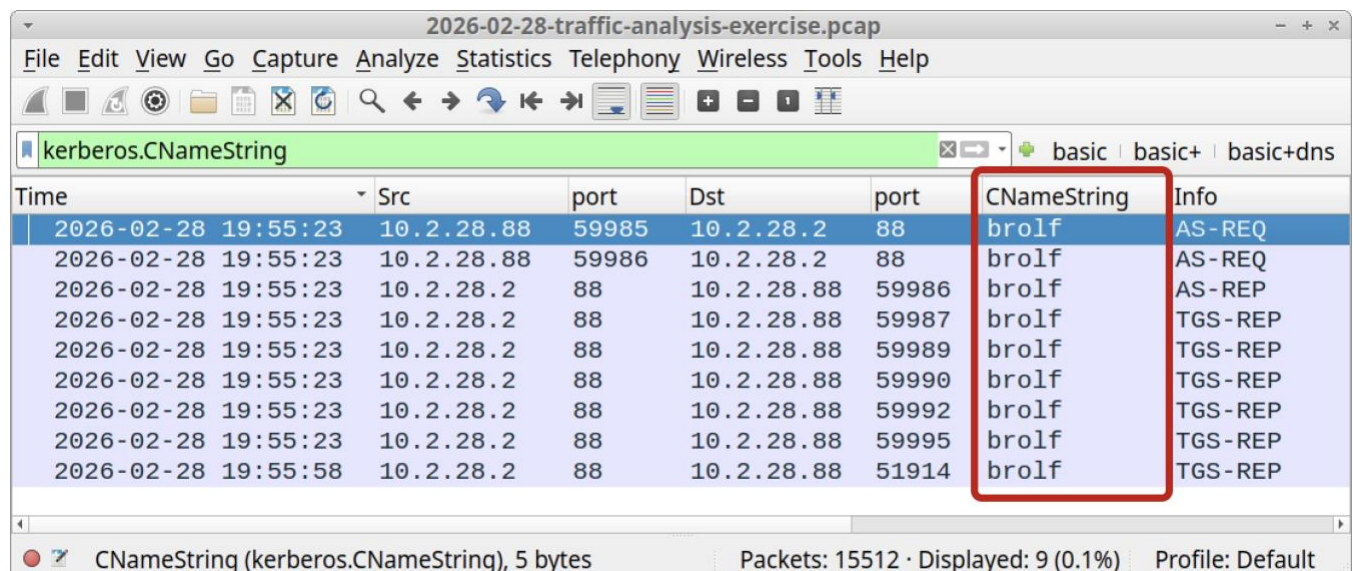
Time	Src	port
2026-02-28 19:55:23	10.2.28.88	59985
2026-02-28 19:55:23	10.2.28.88	59986
2026-02-28 19:55:23	10.2.28.2	88
2026-02-28 19:55:23	10.2.28.2	88
2026-02-28 19:55:23	10.2.28.2	88
2026-02-28 19:55:23	10.2.28.2	88
2026-02-28 19:55:23	10.2.28.2	88
2026-02-28 19:55:23	10.2.28.2	88

[2 Reassembled TCP Segments (1560 bytes): #
Kerberos
Record Mark: 1556 bytes
tgs-rep
pvno: 5
msg-type: krb-tgs-rep (13)
crealm: EASYAS123.TECH
cname
name-type: KRB5-NT-PRINCIPAL (1)
cname-string: 1 item
CNameString: bro1f

Shown above: Filtering to find the `Kerberos.CNameString` value for the user account name.

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We can apply that `CNameString` field as a column, so we don't have to dig into the frame details each time we want to find the user account name in a pcap.



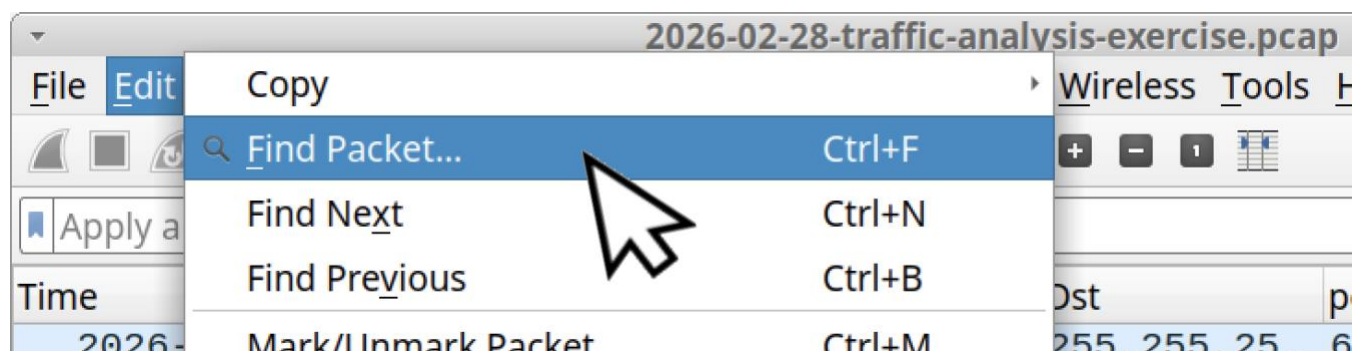
The screenshot shows the Wireshark interface with the packet list table. The 'CNameString' column is highlighted with a red box. The filter 'kerberos.CNameString' is applied. The status bar at the bottom indicates 'CNameString (kerberos.CNameString), 5 bytes' and 'Packets: 15512 · Displayed: 9 (0.1%)'.

Time	Src	port	Dst	port	CNameString	Info
2026-02-28 19:55:23	10.2.28.88	59985	10.2.28.2	88	brolf	AS-REQ
2026-02-28 19:55:23	10.2.28.88	59986	10.2.28.2	88	brolf	AS-REQ
2026-02-28 19:55:23	10.2.28.2	88	10.2.28.88	59986	brolf	AS-REP
2026-02-28 19:55:23	10.2.28.2	88	10.2.28.88	59987	brolf	TGS-REP
2026-02-28 19:55:23	10.2.28.2	88	10.2.28.88	59989	brolf	TGS-REP
2026-02-28 19:55:23	10.2.28.2	88	10.2.28.88	59990	brolf	TGS-REP
2026-02-28 19:55:23	10.2.28.2	88	10.2.28.88	59992	brolf	TGS-REP
2026-02-28 19:55:23	10.2.28.2	88	10.2.28.88	59995	brolf	TGS-REP
2026-02-28 19:55:58	10.2.28.2	88	10.2.28.88	51914	brolf	TGS-REP

Shown above: The `CNameString` value as a column in our Wireshark display.

Now that we have the user account name, it looks like a first initial last name value, in this case, `b` for the first initial and `rolf` for the last name.

We can search the packet details for the case-sensitive string `Rolf` by using the **Find Packet...** function.



Shown above: The **Find Packet...** function from the **Edit** menu.

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The screenshot shows the Wireshark interface for the file '2026-02-28-traffic-analysis-exercise.pcap'. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. Below the menu is a toolbar with various icons. The main window is divided into three panes. The top pane shows a list of packets with columns for Time, Src, port, Dst, port, CNameString, and Info. The second pane shows the details of the selected packet (packet 4), with the 'Full Name' field expanded. The third pane shows the packet bytes. Red arrows point to the search bar, the 'Case sensitive' checkbox, the 'String' search type dropdown, the 'Rolf' search term, and the 'Find' button. A mouse cursor is hovering over the 'Info' column header. The status bar at the bottom indicates 'Full Name (samr.samr_Us...21.full_name), 20 bytes' and 'Packets: 15512 · Displayed: 15512 (100.0%)'.

Time	Src	port	Dst	port	CNameString	Info
2026-02-28 19:55:23	10.2.28.88	59988	10.2.28.2	445		Open
2026-02-28 19:55:23	10.2.28.2	445	10.2.28.88	59988		OpenUs
2026-02-28 19:55:23	10.2.28.88	59988	10.2.28.2	445		QueryU
2026-02-28 19:55:23	10.2.28.2	445	10.2.28.88	59988		QueryU
2026-02-28 19:55:23	10.2.28.88	59988	10.2.28.2	445		QueryS
2026-02-28 19:55:23	10.2.28.2	445	10.2.28.88	59988		QueryS
2026-02-28 19:55:23	10.2.28.88	59988	10.2.28.2	445		GetGro
2026-02-28 19:55:23	10.2.28.2	445	10.2.28.88	59988		GetGro

Full Name
Referent ID: 0x00000000000020000
NDR-Padding: 000000000000
Max Count: 10
Offset: 0
Actual Count: 10
Full Name: Becka Rolf

Home Directory:
Home Drive:
Logon Script:
Profile Path:

Full Name (samr.samr_Us...21.full_name), 20 bytes Packets: 15512 · Displayed: 15512 (100.0%) Profile: Default

Shown above: Searching packet details to find Gabriel Wyatt as the Full Name value.

That all I have for this month's exercise. Catch you all next time!